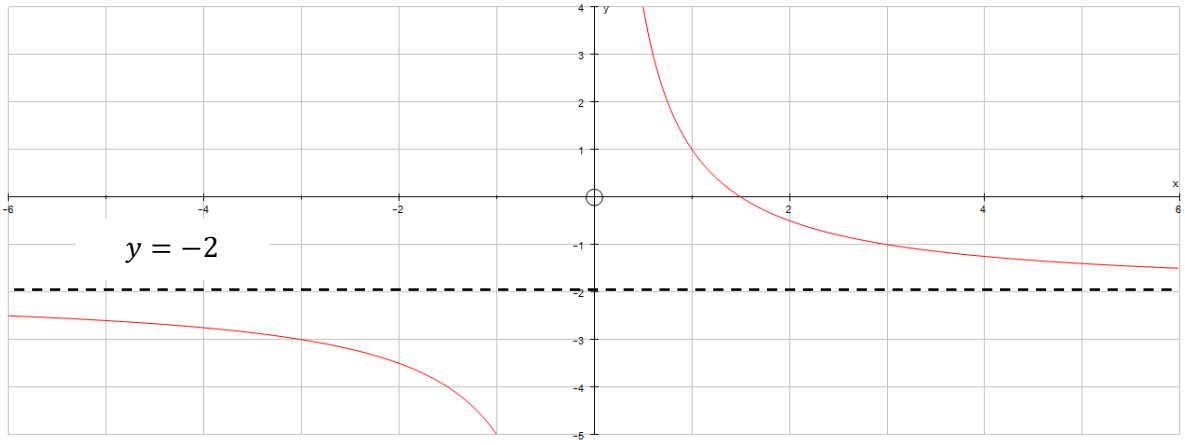
E-classroom	**Name: Memorandum**					
	Gr 10		**Date:**		**Time:**	$2\frac{1}{2}$ **HR**
CAPS Reference	**Functions**					
Topic	**Hyperbolic Function**					

2.1 $f(x) = \frac{3}{x} - 2$

Asymptotes

$x = 0 \quad and \quad y = -2$

x	-3	-2	-1	1	2	3
$\frac{3}{x} - 2$	-3	$-3\frac{1}{2}$	-5	1	$-\frac{1}{2}$	-1



a) $y \in (-\infty; \infty) \; and \; y \neq -2$

b) $y = -2$

c) $x > 1\frac{1}{2} \; and \; x < 0 \; but \; x \neq 0$

d) $h(x) = f(x) + 2$

$= \frac{3}{x} - 2 + 2$

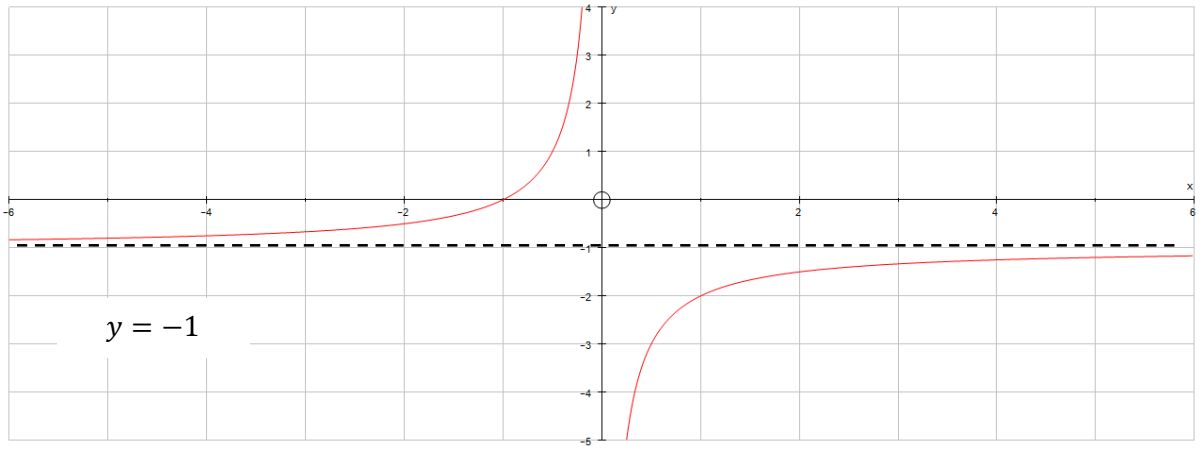
$= \frac{3}{x}$

2.2 $g(x) = -\frac{1}{x} - 1$

Asymptotes

$x = 0 \quad and \quad y = -1$

x	-3	-2	-1	1	2	3
$-\frac{1}{x} - 1$	$-\frac{2}{3}$	$-\frac{1}{2}$	0	-2	$-1\frac{1}{2}$	$-1\frac{1}{3}$



a) $y \in (-\infty; \infty) \; and \; y \neq -1$
b) $y = -1$
c) $x < -1 \; and \; x > 0 \; but \; x \neq 0$
d) $h(x) = -g(x)$

$$= -\left(-\frac{1}{x} - 1\right)$$

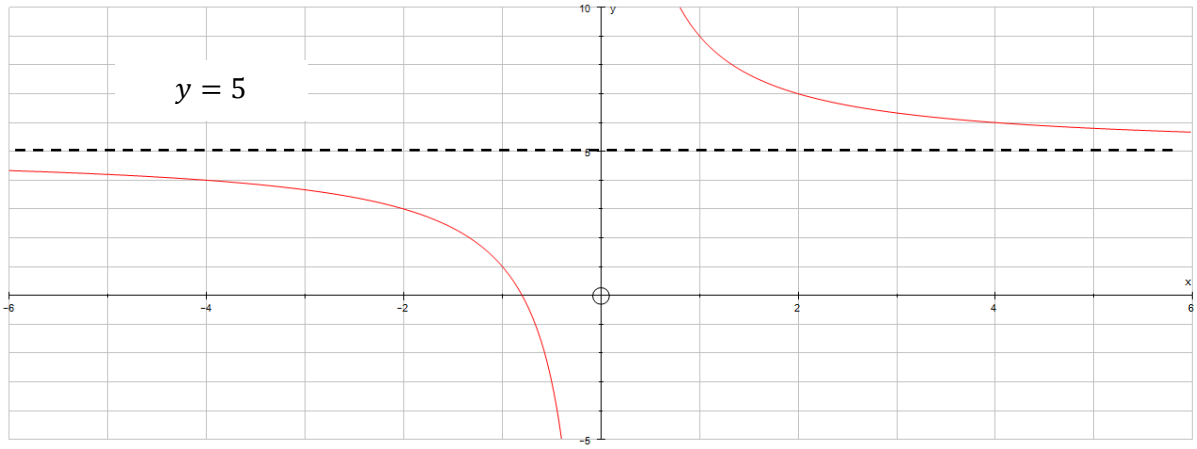
$$= \frac{1}{x} + 1$$

2.3 $f(x) = \frac{4}{x} + 5$

Asymptotes

$x = 0 \qquad and \qquad y = 5$

x	-3	-2	-1	1	2	3
$\frac{4}{x} + 5$	$3\frac{2}{3}$	3	1	9	7	$6\frac{1}{3}$



a) $y \in (-\infty; \infty)\ and\ y \neq 5$

b) $y = 5$

c) $x < -\frac{4}{5}\ and\ x > 0\ but\ x \neq 0$

d) $h(x) = f(x) - 6$

$= \frac{4}{x} + 5 - 6$

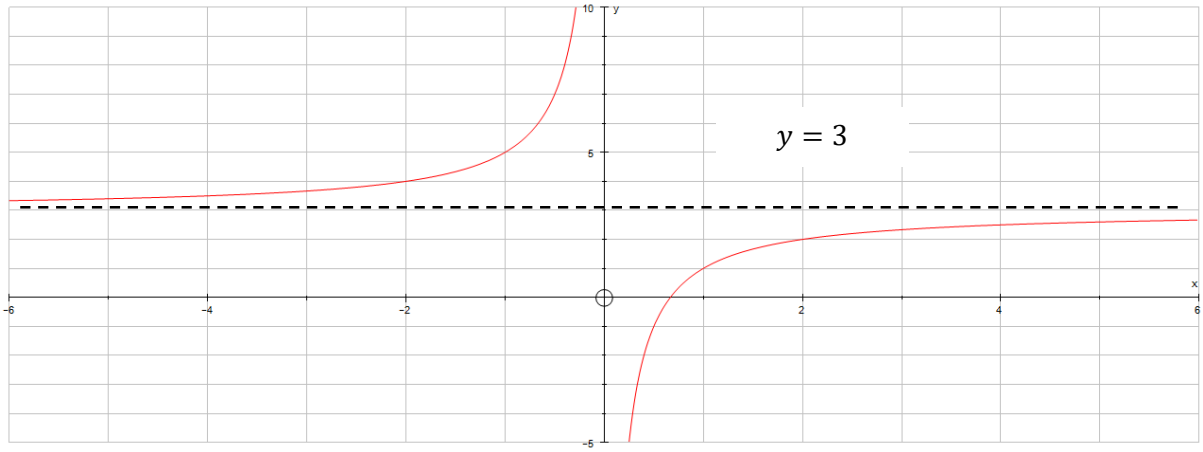
$= \frac{4}{x} - 1$

2.4 $g(x) = -\frac{2}{x} + 3$

Asymptotes

$x = 0 \qquad and \qquad y = 3$

x	-3	-2	-1	1	2	3
$-\frac{2}{x} + 3$	$3\frac{2}{3}$	4	5	1	2	$2\frac{1}{3}$



a) $y \in (-\infty; \infty)\ and\ y \neq 3$

b) $y = 3$

c) $x > \frac{2}{3}\ and\ x < 0\ but\ x \neq 0$

d) $h(x) = -g(x) + 3$

$$= -\left(-\frac{2}{x} + 3\right) + 3$$

$$= \frac{2}{x} - 3 + 3$$

$$= \frac{2}{x}$$

4.1

Step 1: Substitute the asymptotes $y = -2$

$$y = \frac{a}{x} + q$$

$$y = \frac{a}{x} - 2$$

Step 2: Substitute the given coordinate $(1; 0)$

$$y = \frac{a}{x} - 2$$

$$0 = \frac{a}{1} - 2$$

$$2 = 1a$$

$$2 = a$$

Step 3: Substitute a in to the equation.

$$\boldsymbol{f(x) = \frac{2}{x} - 2}$$

4.2

Step 1: Substitute the asymptotes $y = 3$

$$y = \frac{a}{x} + q$$

$$y = \frac{a}{x} + 3$$

Step 2: Substitute the given coordinate $(-4; 1\frac{1}{4})$

$$y = \frac{a}{x} + 3$$

$$1\frac{1}{4} = \frac{a}{-4} + 3$$

$$1\frac{1}{4} - 3 = \frac{a}{-4}$$

$$\frac{7}{4} \times 4 = a$$

$$7 = a$$

Step 3: Substitute a in to the equation.

$$\boldsymbol{f(x) = \frac{7}{x} + 3}$$

4.3

Step 1: Substitute the asymptotes $y = -\frac{1}{2}$

$$y = \frac{a}{x} + q$$

$$y = \frac{a}{x} - \frac{1}{2}$$

Step 2: Substitute the given coordinate $(1; \frac{1}{2})$

$$y = \frac{a}{x} - \frac{1}{2}$$

$$\frac{1}{2} = \frac{a}{1} - \frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2} = a$$

$$1 = a$$

Step 3: Substitute a in to the equation.

$$\boldsymbol{f(x) = \frac{1}{x} - \frac{1}{2}}$$

4.4

Step 1: Substitute the asymptotes $y = 0$

$$y = \frac{a}{x} + q$$

$$y = \frac{a}{x} + 0$$

Step 2: Substitute the given coordinate $(-1; 3)$

$$y = \frac{a}{x}$$

$$3 = \frac{a}{-1}$$

$$3 = -a$$

$$-3 = a$$

Step 3: Substitute a in to the equation.

$$\boldsymbol{f(x) = -\frac{3}{x}}$$